

Guide

The final project for DSGN 348 RP will be custom phone case utilizing multi-material additive manufacturing. Each student is required to design a custom phone case for their personal phone, which is capable of fitting snug & tight, while demonstrating the abilities of additive manufacturing in design. The phone case will be created using AM machines, which will create prototypes for functional testing, ergonomic study, fit/assembly test, visual aid for engineering, and customer feedback. Each student will be required to gather basic user feedback from potential customers & consumers, and use this information to modify and iterate on their design. For this project, we will be using the Stratasys Connex 350 and Formlabs Form2 printers.

Note: Demand for all RP machines is expected to be high through the last 1-2 weeks of this course. Plan accordingly to allow maximum time for your parts to be built.

Part Requirements:

Size Limit: Appropriate size for your phone, with no more than 0.5" addition in any single direction. Pack build files should utilize all available space, so multiple parts should be built in a single tray.

CAD Model: NX or Solidworks may be used for your part modeling. You must create all geometry yourself – no downloaded geometry is permitted to be used for your print. You may use publicly available CAD models to use as reference in creating your models.

Materials: For your final models, you may use either the Connex or Form2 machines to print. A class decision will be made regarding available materials on the Connex.

Other than these basic requirements, you can make any parts you wish!

Iterative Manufacturing: Each student will generate a minimum of three printed parts.

1st model: Size & feature study, initial fits

2nd and 3rd model: Materials study, user feedback, fit tests.

You will be required to operate each piece of equipment with your classmates, and schedule build time in conjunction with other students. The following must be executed on your own:

- * Creating suitable build files, assign materials to be used, and hollow/shell to minimize material usage
- * Setup the machine to print, and address any material / consumable replacements if needed

- * Create a pack build using all available space. Multiple students should submit a job at once
- * Clean the machine once finished per outlined instructions
- * Remove support, and polish / clean to final presentation look.

If you have ANY questions regarding machine operation, contact the course instructor/TA for help.

User feedback: After rapid prototyping your 2nd overall model, user feedback must be gathered from potential consumers and users outside of this course. Please gather feedback from at least 5 different people regarding a certain aspect of your product which sees heavy interaction with the consumer. For example, a handle on a door would be an area to subject to user testing. Specifics as to the type of user feedback you will be required to generate will be reviewed in class.

You will be required to document the questions and setup you are using to generate user feedback, as well as the answers and comments you receive on the product. This documentation will be required as an appendix within your final project report.

Project Writeup: A final document, which will present the various stages of your part development, will be required of each student. There is a soft limit of ~10 pages. Your report should include the following sections at minimum:

- * Development of your model: Hand sketches, CAD Models, and Model pictures showing the development of your model, and how AM was able to help create your part. Emphasize the use of rapid prototyping within your design.
- * Review of three iterated designs. Highlight changes from one version to the next, including explanations as to why changes were made.
- * Overview your user feedback. What areas of your product did you choose to improve upon based on this feedback? What were the results? How did you change your model? You may include this in the previous section of iterated designs.
- * Overall lessons learned. What did you learn when modeling / reverse engineering surfaces in CAD? Outline any problems you had during the rapid prototyping process, and how you solved them. What do you see as the main benefits of using this rapid product development process?

Due Date: Submission of all parts, report, CAD models, and scanned surface files will be announced on the course syllabus, and on Canvas.

All files and write-up should be submitted via Canvas, with parts and assembly files in a single zip file. Final parts are due to the instructor by the due date and time.

This is a comprehensive project that blends high-fidelity engineering with user-centric design. Since you are using the **Stratasys Connex 350** (PolyJet) and **Formlabs Form2** (SLA), your report should lean heavily into the advantages of those specific technologies—namely multi-material gradients and high surface finish.

Here is a structured outline for your **DSGN 348 Final Report** to ensure you hit all the rubric requirements within the ~10-page soft limit.

Final Report Outline: Custom Multi-Material Phone Case

1. Introduction & Design Intent

- **Project Objective:** Brief overview of the goal—creating a custom, multi-material phone case using AM.
- **Phone Specification:** Identify the phone model used for the "snug & tight" fit requirement.
- **Design Philosophy:** What was your goal? (e.g., impact resistance, unique aesthetics, ergonomic grip, or integrated mechanical features).

2. Development & Conceptualization

- **Initial Ideation:** Include scans or photos of original hand sketches.
- **CAD Strategy:** * Discuss the workflow in **SolidWorks** or **NX**.
 - Explain how you handled the "reference geometry" (e.g., using a public model of the phone to create the negative cavity).
 - **AM Design Considerations:** Detail how you designed for hollowing/shelling to save material and how you assigned different material bodies for the Connex 350.

3. Iterative Design Review (The Three Models)

This section is the core of your technical narrative.

Model 1: Size, Feature & Initial Fit Study

- **Goal:** Validating the primary dimensions and "snugness."
- **Outcome:** Photos of the print and the phone partially/fully inserted.
- **Refinement:** What was wrong? (e.g., "The button cutouts were misaligned by 1mm" or "The snap-fit was too tight").

Model 2: Ergonomics & Material Study

- **Goal:** Testing the "feel" and multi-material integration.
- **AM Process:** Discuss using the Form2 or Connex for specific textures.
- **User Interaction Point:** Define the specific area you chose for testing (e.g., the volume rocker feel or the corner grip).

Model 3: Final Optimized Design

- **Goal:** Final assembly, polished aesthetics, and resolved user issues.
- **Final Result:** High-quality photos of the finished, cleaned, and polished case.

4. User Feedback & Analysis

- **Testing Methodology:** Describe the setup. How did you present Model 2 to your 5 participants?
- **Documented Feedback:** Summarize the raw data (detailed charts or lists should go in the Appendix).
- **Design Response:** "Users felt the TPU-like material on the edges was too slippery, so I added a knurled texture in the final CAD."

5. Manufacturing & Workflow Reflection

- **Build Prep:** Discuss creating the pack builds and maximizing tray space with classmates.
- **Machine Operation:** * **Connex 350:** Experiences with resin/support management.
 - **Form2:** Experiences with IPA washing and UV curing.
- **Post-Processing:** Describe the removal of support material and the polishing steps taken for the "presentation look."

6. Lessons Learned & Conclusion

- **Surface Modeling:** What were the challenges of reverse-engineering the phone's curves?
- **Problem Solving:** Identify a specific AM failure (e.g., a warped thin wall or clogged nozzle) and how you pivoted.
- **The Value of RP:** Reflect on how having a physical part in hand every week changed your design compared to just looking at a screen.

7. Appendices

- **Appendix A:** Full User Feedback Questionnaire and raw responses.

- **Appendix B:** Technical Drawings/Exploded Views.
- **Appendix C:** High-resolution screenshots of the final "Pack Build" tray layout.

Pro-Tips for the DSGN 348 Report:

- **Highlight the "Multi-Material" aspect:** Since you have access to the Connex, make sure to explain *why* you chose specific Shore-A hardness values (if using Digital Materials) for certain parts of the case (e.g., soft bumpers vs. a hard shell).
- **Show, Don't Just Tell:** Use side-by-side photos of the CAD model next to the physical printed part to demonstrate your "visual aid for engineering."
- **Watch the Page Count:** Use the "Iterative Design" section to house your best photos so they serve both as evidence of work and visual breaks in the text.

Writeup

Google Pixel 7 Phone Case Design & Manufacturing

DSGN 348: Rapid Prototyping

Danish Tiro Galebotswe

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Introduction & Design Intent

Project Objective

This project for DSGN 348 RP involved the design of a custom phone case utilizing multi-material additive manufacturing. The goal was to develop a design for my personal phone with a snug & tight fitting which would be manufactured using additive manufacturing methods particularly the Form4 Lab SLA printers, Pruse MK4/MK4s, and Stratasys J826 (which later had a breakdown). Different phone case prototypes would undergo functional testing, ergonomic study, fit/assembly tests, visual aid for engineering and customer feedback.

Phone Specification

The phone model used for this project was a Google Pixel 7 which has the following dimensions: 6.13 x 2.88 x 0.34 in (155.6 x 73.2 x 8.7 mm).

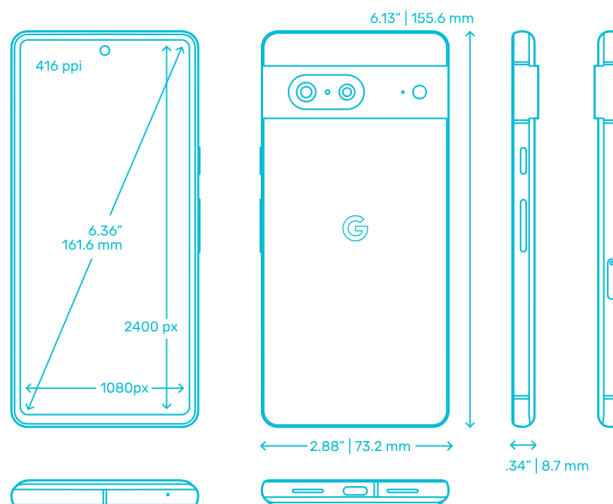


Figure XX: Dimensions of the Google Pixel 7. ¹

Design Philosophy

In addition to the goal of designing and manufacturing a snug and fit phone case, the philosophy was to develop a phone case that exhibited the beauty of additive manufacturing and engineering. This was to be achieved by adding on intricate details to the phone case which would be difficult to achieve with conventional manufacturing methods like injection molding with limited time available. Therefore, the first iteration involved a map of Botswana (sideways it looked like Texas) and later on involved a snap-on optimized axial fan with the words MSAM standing for Master of Science in Advanced Manufacturing.

¹ "Google Pixel 7," Dimensions.com, accessed May 8, 2026, <https://www.dimensions.com/element/google-pixel-7>.

Development & Conceptualization

Initial ideas for the phone case were influenced by my country of origin: Botswana. The zebra animal is the country's national animal hence the first and second idea involved placing the face of a zebra on the back of the phone case.

Initial Ideation

Idea #1

This idea involved using the capabilities of the Stratasys J826 Polyjet Machine. The zebra head would be imbedded into the back of the phone case with a different color and texture to the rest of the phone case.

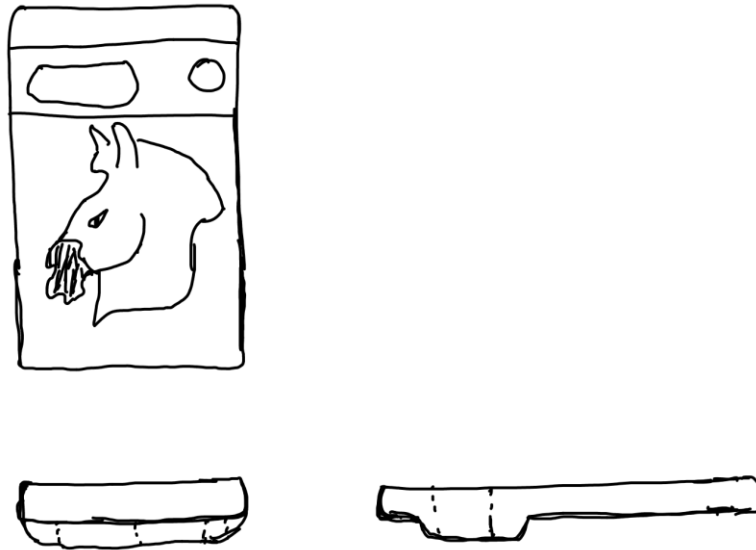


Figure XX: Idea #1 sketches

Idea #2

This idea involved having the zebra head as an extrusion on the back of the phone case. The AM method that would be used for this idea would be FDM with the Prusa MK4/4S or SLA with Form 4/4B Resin printers.

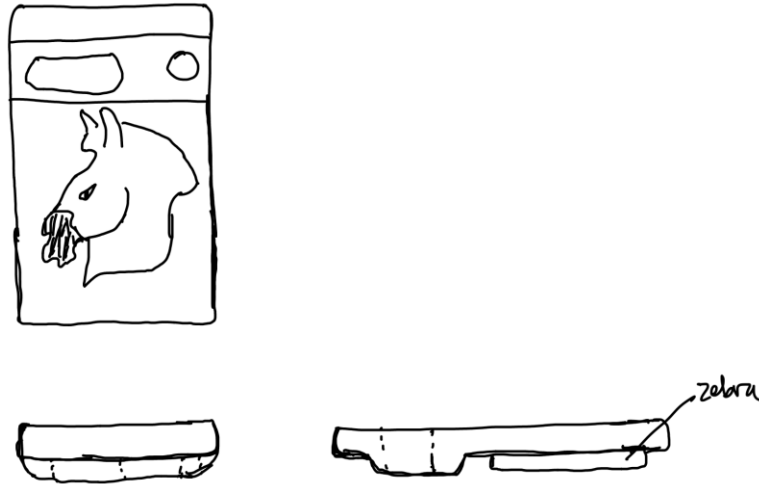


Figure XX: Idea #2 sketches.

Idea #3

This idea was more focused on the functionality of the phone case with an opening for human fingers (except thumb) to hold the phone.

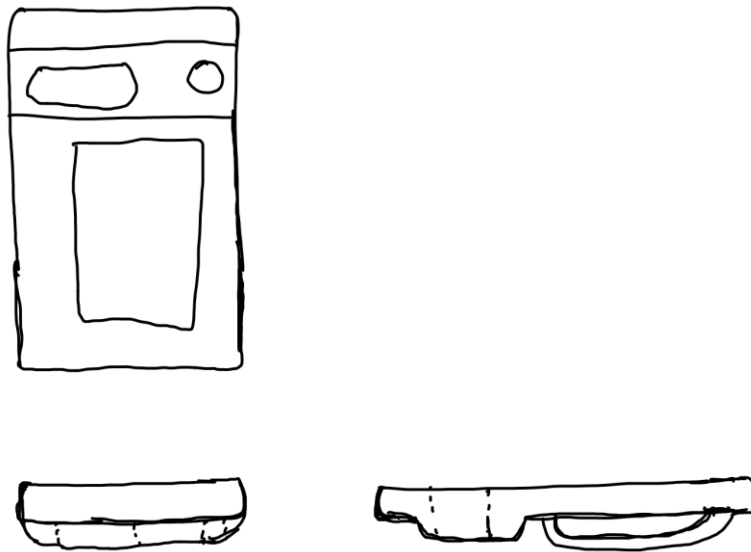


Figure XX: Idea #3 sketches.

These ideas were developed as there were considered to be less challenging therefore, the design pivoted to an extrusion of the outline of the country of Botswana on the back of the phone case with the letters M-S-A-M. This was used adopted for the first prototype.

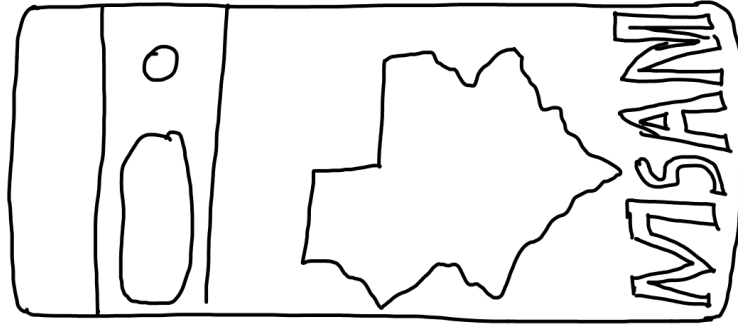


Figure XX: Idea #4 sketch.

For the second and third prototype, the design evolved by adding an axial fan blade on the back of the phone case that would be a snap-on. Furthermore, the letters were kept at the back that would show the capabilities of Additive Manufacturing in achieving intricate and complex designs.

CAD Strategy

The phone case 3D model was developed in Solidworks using surface modeling toolbox. Initially, a caliper was used to reverse engineer the following dimensions of the phone:

- Overall height, width and thickness
- Camera extrusion thickness
- Camera Lens width, length and position
- Flashlight diameter
- Power Button length, width and position
- USB and Speakers length, width and position

The following surface modeling operations were used:

1. Sweep Surface
2. Surface fill
3. Knit Surface
4. Thicken
5. Extruded cut (solid modeling)
6. Revolve extrude (solid modeling)

Iterative Design Review

Model 1: Size, Feature & Initial Fit Study



Figure XX: Model 1 made out of TPU on the Prusa Mk4/4S

The goal of this model was to validate the the primary dimensions especially the overall length, thickness and width as well as the cutouts for the USB, speak, camera and the power button. While the cutouts, overall length, and width dimension came out good, the thickness of the phone case which was 1mm was too flexible with TPU so the snap-fit was weak. An improvement on the next design was to keep the same thickness but use a different material Tough 1500 using Form 4/4B SLA printers.

Model 2: Ergonomics & Material Study



Figure XX: (Left) Model 2 V1 made out of Tough 1500 on the Form 4/4B SLA. (Right) Model 2 V2 made out of Tough 2000.

The goal of this model was to test the "feel" of the phone case and obtain feedback from five (5) different would-be users of the phone case. Users identified the following major issues:

- Rough finish of the phone case felt by the bumps: four out of the five interviewees expressed a dislike for this rough finish
- "Unsatisfying" power button: three out of the five users explained how they applied a lot more force than they have experienced before
- Axial Fan mounted too low position-wise and gap to the phone: all the interviewees expressed a strong dislike for how low the snap location for the axial fan which impeded holding the phone

while trying to fidget with it. Furthermore, they expressed how it was uncomfortable to pass their fingers through the gap between the fan blades and the back of the phone case.

This feedback was adopted and changes were made to the design likewise. The axial fan position which was the biggest pain point was moved up and modifications made to the fan. Another significant change was to use different materials for the phone case (Tough 1500) and the fan (Tough 2000). This was due to how the fan blade trailing edges were weak from Tough 1500.

Model 3: Final Design



Figure XX: Final Design of the phone case and the fan.

The figure above shows the final design of the phone case and the fan. It was easy to fit fingers between the fan blades and the back of the phone case so more comfortable holding the phone while the blades are on.

Lessons Learned & Conclusion

Lessons Learned

This project provided valuable experience in iterative product design, additive manufacturing process selection, and user-centered engineering design. While the initial goal was to create a snug-fitting custom phone case for the Google Pixel 7, the project evolved into a broader study of how geometry, material properties, manufacturing method, and user ergonomics interact to determine final product success.

One of the primary lessons learned was the importance of material selection in relation to functional requirements. The first TPU prototype demonstrated that even when dimensional accuracy is acceptable, insufficient stiffness can result in weak snap-fit retention and poor structural performance. This highlighted that successful design is not solely dependent on geometry, but on matching material mechanical properties such as stiffness, toughness, and flexibility, to the intended application. Later iterations using Tough 1500 and Tough 2000 resin showed improved rigidity, better retention, and greater durability for thinner and more intricate features.

Another major takeaway was the value of rapid prototyping and iterative testing. By producing multiple physical prototypes rather than relying solely on CAD evaluation, issues related to ergonomics, tactile feel, and usability were quickly identified. User feedback revealed problems that were not immediately obvious during digital design, including the rough surface finish, high actuation force required for the power button, and poor placement of the snap-on axial fan feature. These findings reinforced that engineering design must consider human interaction and product experience, not only dimensional fit and manufacturability.

The project also deepened understanding of design for additive manufacturing (DfAM). Additive manufacturing enabled the incorporation of intricate details such as the Botswana outline, embossed lettering, and snap-on fan geometry that would be significantly more difficult or costly to manufacture conventionally. At the same time, it became clear that additive manufacturing introduces its own design constraints, including surface finish considerations, anisotropic strength behavior, minimum feature thickness requirements, and material-process compatibility. Designing successfully for AM requires balancing creative geometric freedom with process limitations.

Finally, the project emphasized the importance of engineering flexibility. Several pivots were required throughout development from changes in concept direction to redesigns driven by material performance, user feedback, and equipment limitations such as the Stratasys J826 breakdown. Adapting quickly while maintaining design intent was a critical part of the development process and closely mirrors real-world engineering practice.

Conclusion

Overall, the Google Pixel 7 phone case project successfully met its objective of designing and manufacturing a custom, functional, and visually distinctive phone case optimized for additive manufacturing. Through three major design iterations, the final product achieved a snug fit, improved ergonomic comfort, enhanced tactile interaction, and stronger snap-on accessory functionality. The final design also fulfilled the original design philosophy of showcasing the capabilities of additive manufacturing through complex geometry and personalized aesthetic features.

Beyond the final product itself, the greatest success of this project was the development of a structured engineering workflow involving concept generation, CAD modeling, prototyping, testing, feedback incorporation, and redesign. This project demonstrated that effective engineering design is inherently iterative, and that the best solutions emerge through repeated evaluation and refinement rather than initial perfection.

In conclusion, this project successfully combined engineering analysis, manufacturing knowledge, and creative design, resulting in both a functional end product and a deeper understanding of additive manufacturing-driven product development.

Appendices

Appendix A: Full User Feedback Questionnaire and Raw Responses

All five users were asked the following questions:

1. How much do you like the material and its flexibility?
2. How do the edges feel?
3. How easy or difficult was it to insert the phone and take it out of the phone case? 1 (super easy) - 5 (super hard) ranking
4. Did you like the axial fan accessory? How easy or difficult was it to snap on and off the back of the phone?
5. Was the power button comfortable and satisfying to push?

User #1 Feedback

1. The material feels cool and I like it. It is more flexible than I thought. The USB cutout is really warped and bumpy though
2. Corners are not rounded and are sharp.
3. It was kinda difficult to insert the phone into the case but easy to take it out
4. The axial fan is pretty cool. Its like a good fidget. But I would like a phone case with a stand instead. Also I don't know how I would store this?
5. The power button is not satisfying and I can't feel the pressing on the button

User #2 Feedback

1. The material is smooth, uniform. The inside is spiky though
2. The edges of the screen have spiky dots that are sharp. But the rest of it is good.
3. I would give it a 2 for inserting and removing the phone from the phone case. But the outer lip of the phone case does not go over the screen.
4. I am not interested in the attachment. Also the blades are sharp and spikey. The attachment seems a little floppy but it can snap on and off multiple cycles.
5. It was difficult to if you pressed the button. There is no travel with the button.

User #3 Feedback

1. The material feels nice, solid and stiff which is good. But there are bumps and spikes inside. The phone case is not too stiff but not too flexible.
2. The edges like rounded sharp edges. Its little non rounded and more comfortable
3. I would rank the inserting process with a 4.5. It required a lot of pushing in. It was better to push in from the side. Taking off was easier.
4. I like spinning things and fidgets. Move the blade up to have more space and also increase the gap between the fan and the back of the phone. Difficult to spin while holding the phone. It was easy to snap on and off

5. The power button feels nice. Its a bit rough increase the width. Its thin so its comfortable pressing feeling.

User #4 Feedback

1. I like the texture and the harder feel. I dont like the friction of the material though. Its too smooth so it rolls out my hand. I like the color - off black. I never thought about the flexibility. I dont have an opinion that.
2. The edges are smooth except for the bumps on the some of the them.
3. I would rank the inserting process easy and not difficult. If I try top removal of the phone, it is difficult so I would rank it a 4. But the side removal was very easy like 1.
4. When the blade is on, I like to fidget with it. But I dont think I would carry the fan around. Storage seems like the biggest issue for me. The blades are very fragile. It is very easy to snap on and off but I think a much wider gap between the fan blades and the back of the phone would be better
5. The power button is nice and noticeable. It feels good and easily travels as I press.

User #5 Feedback

1. Its noce an dmore flexible. More bending. The colow looks good. Maybe the material is not good for scratches and finger prints.
2. The edge textures are not bad. There are just bumps of the lower edges. Everywhere else is fine but the bottom is not good
3. Inserting the phone was easy so 1 but the sides were an issue when taking out so 2.
4. [User did not find it intuitive that it would snap on and off] Raise th blade gap on the hands. Is there a way to automate it? I dont want to move it with my fingers. [User tried to balance the phone and phone case together on the blade]
5. Power button is okay but I was using a lot of force to press it. Make something softer.